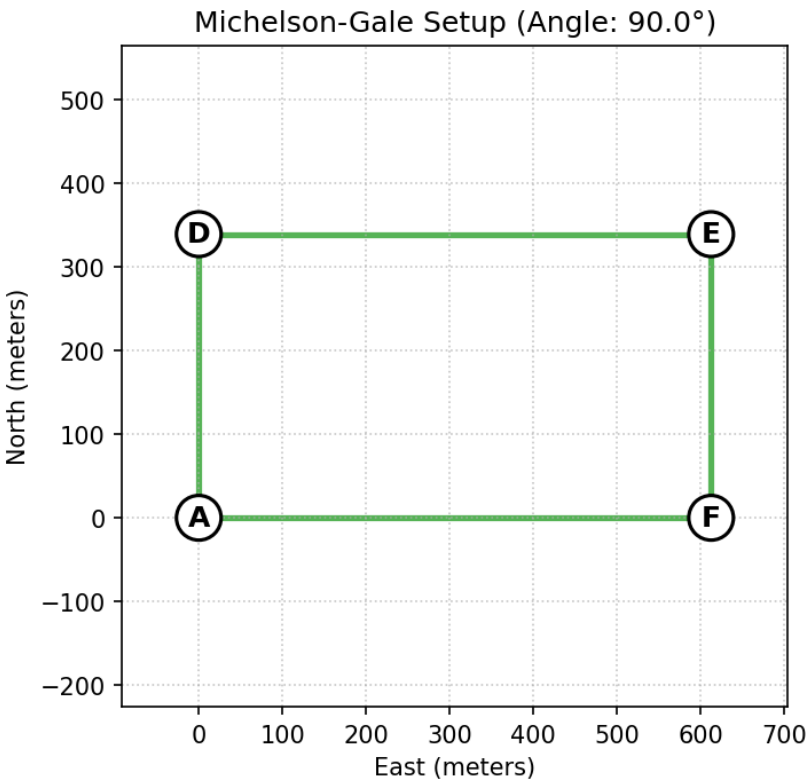


3D ECI Interferometer Simulation Report

SCENARIO 1a: Michelson-Gale: No Earth rotation + No Drift Vector



Location: 41.76666666666667 deg N
Loop Shape: A-D: 339.2424m, A-F: 612.648m @ Angle: 90.0 deg
Wavelength: 570e-9 m
Kinematics: Earth Rot: OFF | Drift Vector: 0.0 m/s @ 0.0 deg

--- Michelson-Gale (0 deg Orientation) ---
Time Diff: 0.0000000000 ns
Distance Diff: 0.0000000000 nm
Fringe Shift: 0.0000000000 fringes

SCENARIO 1b (as expected): Michelson-Gale: Earth rotation + No Drift Vector

Location: 41.76666666666667 deg N
Loop Shape: A-D: 339.2424m, A-F: 612.648m @ Angle: 90.0 deg
Wavelength: 570e-9 m
Kinematics: Earth Rot: ON | Drift Vector: 0.0 m/s @ 0.0 deg

--- Michelson-Gale (0 deg Orientation) ---
Time Diff: 0.0000004493 ns
Distance Diff: 134.6953190521 nm
Fringe Shift: 0.2363075773 fringes

SCENARIO 1c: Michelson-Gale: Earth rotation + Drift Vector

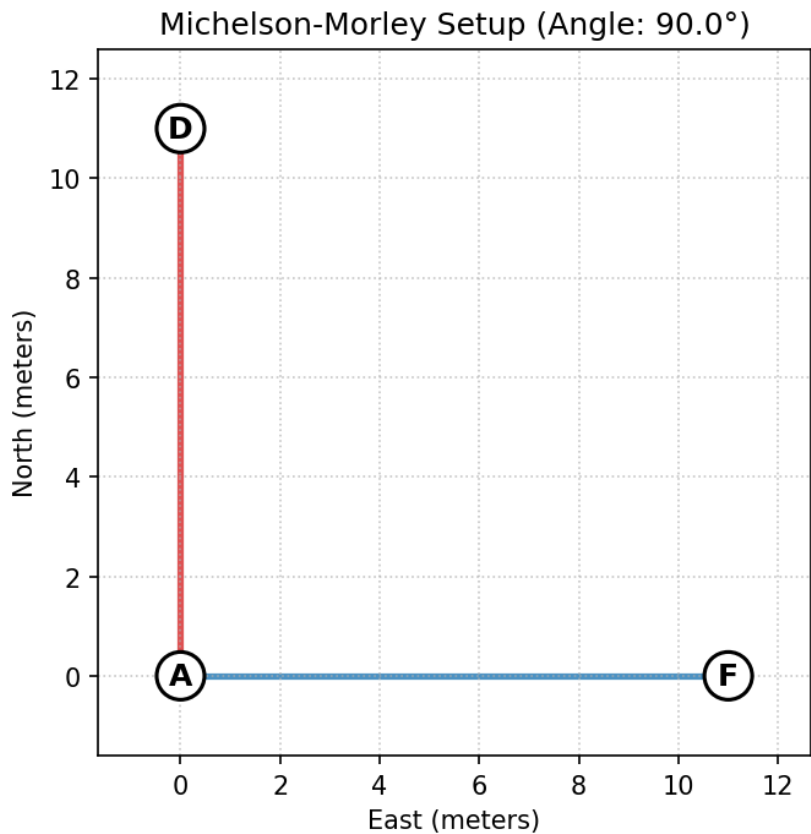
Location: 41.76666666666667 deg N
Loop Shape: A-D: 339.2424m, A-F: 612.648m @ Angle: 90.0 deg

3D ECI Interferometer Simulation Report

Wavelength: 570e-9 m
Kinematics: Earth Rot: ON | Drift Vector: 30000.0 m/s @ 0.0 deg

--- Michelson-Gale (0 deg Orientation) ---
Time Diff: 0.0000004493 ns
Distance Diff: 134.6953211299 nm
Fringe Shift: 0.2363075809 fringes

SCENARIO 2a (as expected): Michelson-Morley: Earth rotation + No Drift Vector



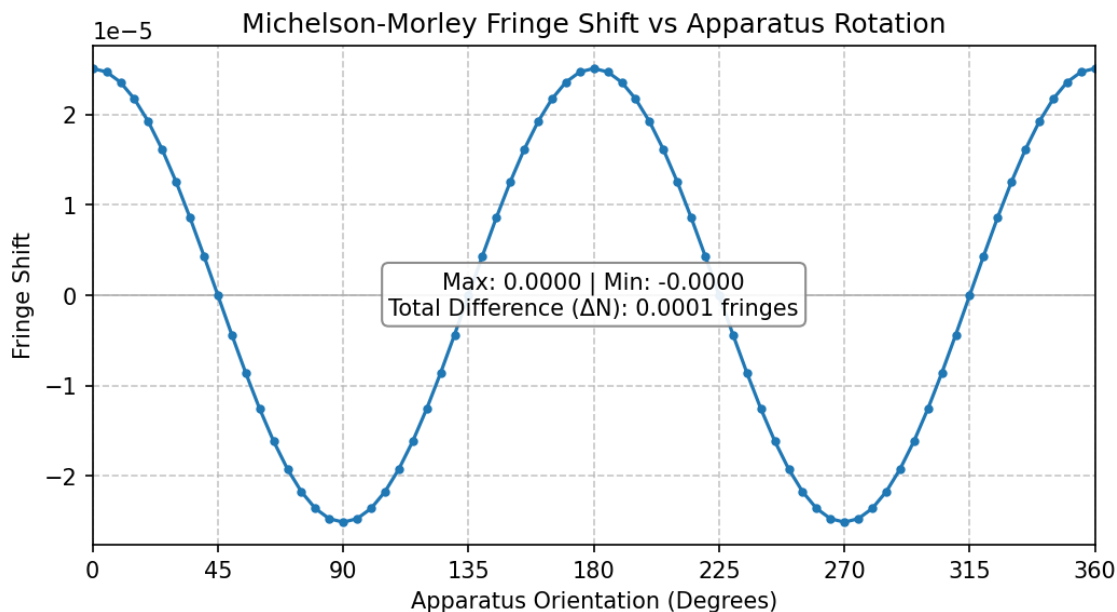
Location: 41.5 deg N
Loop Shape: A-D: 11m, A-F: 11m @ Angle: 90.0 deg
Wavelength: 590e-09 m
Kinematics: Earth Rot: ON | Drift Vector: 0.0 m/s @ 0.0 deg

--- Michelson-Morley (0 deg Orientation) ---
Time Diff: 0.0000000000 ns
Distance Diff: 0.0148179157 nm
Fringe Shift: 0.0000251151 fringes

--- Time Dilation Analysis (ECI Frame) ---
Formulas applied for theoretical comparison:
 $\gamma = 1 / \sqrt{1 - (v/c)^2}$
Transverse Diff % = $(\gamma - 1) * 100$
Longitudinal Diff % = $(\gamma^2 - 1) * 100$

3D ECI Interferometer Simulation Report

Apparatus ECI Velocity: 347.9506 m/s
Arm A-D (Moving vs Stat): 0.000000000067354 %
Arm A-F (Moving vs Stat): 0.000000000134708 %
Theoretical Transverse (γ): 0.000000000067354 %
Theoretical Longitudinal (γ^2): 0.000000000134708 %



SCENARIO 2b: Michelson-Morley: No Earth rotation + Drift Vector

Location: 41.5 deg N
Loop Shape: A-D: 11m, A-F: 11m @ Angle: 90.0 deg
Wavelength: 590e-09 m
Kinematics: Earth Rot: OFF | Drift Vector: 30000.0 m/s @ 0.0 deg

--- Michelson-Morley (0 deg Orientation) ---

Time Diff: 0.0000003674 ns
Distance Diff: 110.1523569281 nm
Fringe Shift: 0.1866989100 fringes

--- Time Dilation Analysis (ECI Frame) ---

Formulas applied for theoretical comparison:

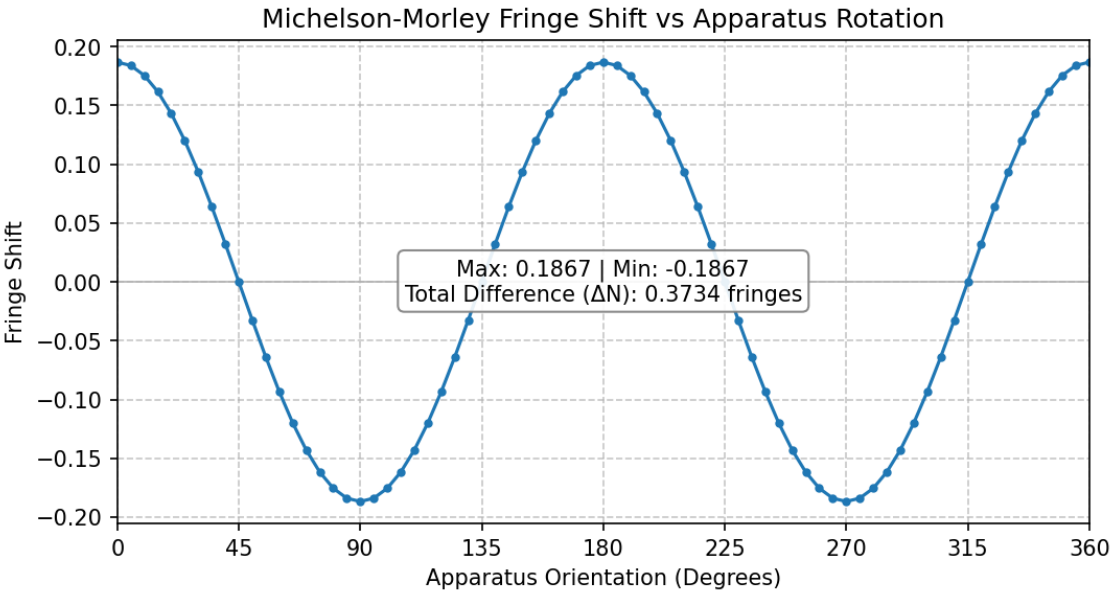
$$\gamma = 1 / \sqrt{1 - (v/c)^2}$$

$$\text{Transverse Diff \%} = (\gamma - 1) * 100$$

$$\text{Longitudinal Diff \%} = (\gamma^2 - 1) * 100$$

Apparatus ECI Velocity: 30000.0000 m/s
Arm A-D (Moving vs Stat): 0.000000500692529 %
Arm A-F (Moving vs Stat): 0.000001001385060 %
Theoretical Transverse (γ): 0.000000500692529 %
Theoretical Longitudinal (γ^2): 0.000001001385060 %

3D ECI Interferometer Simulation Report



SCENARIO 2c: Michelson-Morley: Earth rotation + 30km/s Drift Vector

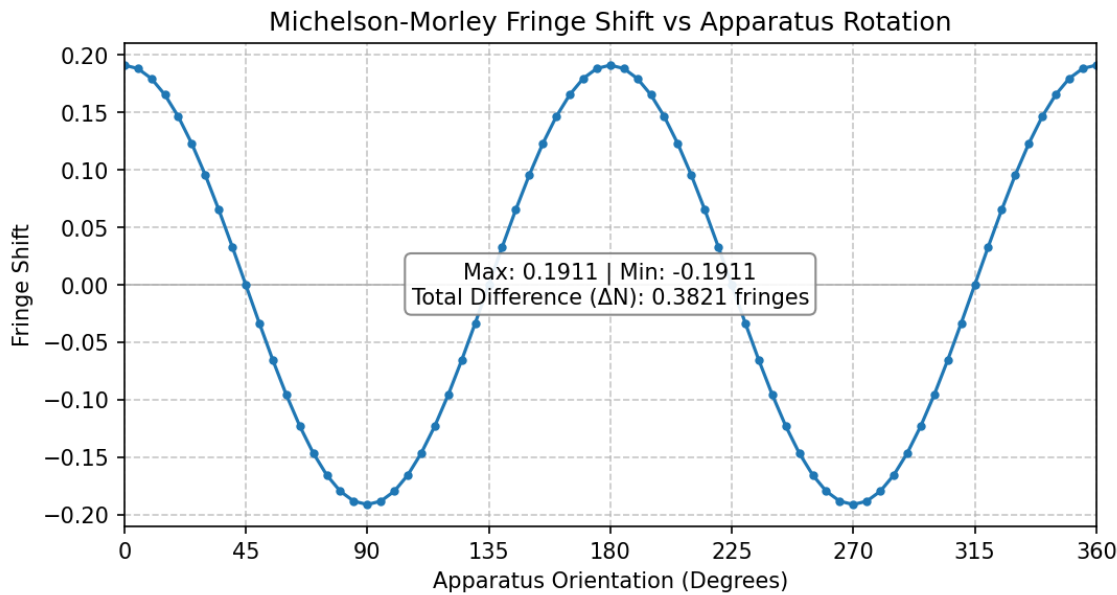
Location: 41.5 deg N
Loop Shape: A-D: 11m, A-F: 11m @ Angle: 90.0 deg
Wavelength: 590e-09 m
Kinematics: Earth Rot: ON | Drift Vector: 30000.0 m/s @ 0.0 deg

--- Michelson-Morley (0 deg Orientation) ---
Time Diff: 0.0000003760 ns
Distance Diff: 112.7223467806 nm
Fringe Shift: 0.1910548251 fringes

--- Time Dilation Analysis (ECI Frame) ---
Formulas applied for theoretical comparison:
 $\gamma = 1 / \sqrt{1 - (v/c)^2}$
Transverse Diff % = $(\gamma - 1) * 100$
Longitudinal Diff % = $(\gamma^2 - 1) * 100$

Apparatus ECI Velocity: 30347.9506 m/s
Arm A-D (Moving vs Stat): 0.000000512374301 %
Arm A-F (Moving vs Stat): 0.000001024748604 %
Theoretical Transverse (γ): 0.000000512374301 %
Theoretical Longitudinal (γ^2): 0.000001024748604 %

3D ECI Interferometer Simulation Report



SCENARIO 4: MM Experiment Moving West on a Train

Location: 41.5 deg N
Loop Shape: A-D: 11m, A-F: 11m @ Angle: 90.0 deg
Wavelength: 590e-09 m
Kinematics: Earth Rot: ON | Drift Vector: 100.0 m/s @ 180.0 deg

--- Michelson-Morley (0 deg Orientation) ---

Time Diff: 0.0000000000 ns
Distance Diff: 0.0075245912 nm
Fringe Shift: 0.0000127535 fringes

--- Time Dilation Analysis (ECI Frame) ---

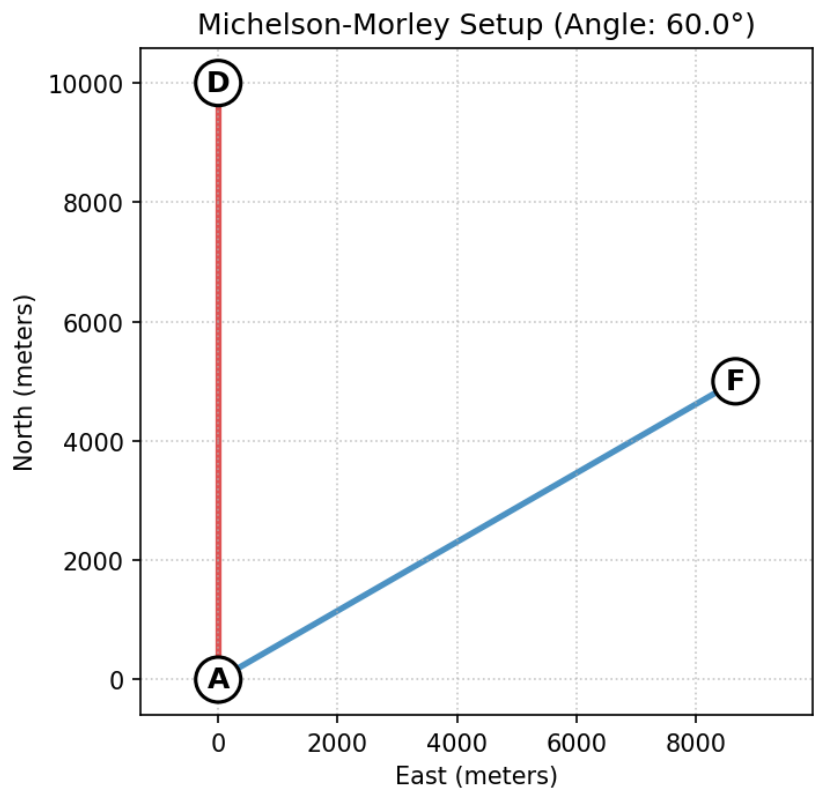
Formulas applied for theoretical comparison:

$\gamma = 1 / \sqrt{1 - (v/c)^2}$
Transverse Diff % = $(\gamma - 1) * 100$
Longitudinal Diff % = $(\gamma^2 - 1) * 100$

Apparatus ECI Velocity: 247.9506 m/s
Arm A-D (Moving vs Stat): 0.000000000034202 %
Arm A-F (Moving vs Stat): 0.000000000068405 %
Theoretical Transverse (γ): 0.000000000034203 %
Theoretical Longitudinal (γ^2): 0.000000000068405 %

SCENARIO 3a: Einstein telescope; Earth doesn't rotate

3D ECI Interferometer Simulation Report



Location: 50.7542 deg N
Loop Shape: A-D: 10000.0m, A-F: 10000.0m @ Angle: 60.0 deg
Wavelength: 1.55e-06 m
Kinematics: Earth Rot: OFF | Drift Vector: 0.0 m/s @ 0.0 deg

--- Michelson-Morley (0 deg Orientation) ---

Time Diff: -0.0000000000 ns
Distance Diff: -0.0000000000 nm
Fringe Shift: -0.0000000000 fringes

SCENARIO 3b (as expected): Einstein telescope; Earth rotates

Location: 50.7542 deg N
Loop Shape: A-D: 10000.0m, A-F: 10000.0m @ Angle: 60.0 deg
Wavelength: 1.55e-06 m
Kinematics: Earth Rot: ON | Drift Vector: 0.0 m/s @ 0.0 deg

--- Michelson-Morley (0 deg Orientation) ---

Time Diff: 0.0000000241 ns
Distance Diff: 7.2319295728 nm
Fringe Shift: 0.0046657610 fringes

SCENARIO 3c (on Equator): Einstein telescope; Earth rotates

Location: 0 deg N
Loop Shape: A-D: 10000.0m, A-F: 10000.0m @ Angle: 60.0 deg
Wavelength: 1.55e-06 m

3D ECI Interferometer Simulation Report

Kinematics: Earth Rot: ON | Drift Vector: 0.0 m/s @ 0.0 deg

--- Michelson-Morley (0 deg Orientation) ---

Time Diff: 0.0000000601 ns

Distance Diff: 18.0111318182 nm

Fringe Shift: 0.0116200850 fringes